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Nobre, T.L: Conceptualization, Writing -Original draft preparation, Writing – Review and Editing, Project administration. **Rocha, L. Y:** Conceptualization, Writing -Original draft preparation. **Carbone, P. O:** Methodology, Formal analysis, Writing -Original draft preparation. **Grassi, M. C. F:** Methodology, Formal analysis. **Júnior, R. O de:** Methodology, Formal analysis. **Caperuto, E. C:** Conceptualization, Writing – Review and Editing, Supervision.

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ABSTRACT

Indirect biological markers of muscle damage are used to measure the level of muscle damage caused by exercise in athletes who are constantly exposed to overloads. However, infrared thermography is an evaluation and diagnostic method that has been frequently used to identify the location of possible injuries caused by temperature asymmetries. A systematic review was carried out with the aim of verifying whether changes in skin temperature can be compared with the inflammatory physiological changes observed after exercise protocols in trained individuals. The PRISMA recommendations were followed in January 2023, in the Pubmed, Medline, Lilacs, Scielo, Sportdiscus and Embase databases. A total of 165 studies were found. After screening, five studies were selected for this systematic review. The results suggest that infrared thermography is being widely used for injury screening, and that few studies associate the method with biomarker tests. However, it may be more advantageous for identifying muscle damage, as it does not determine the location of the injury, but only what is injured. Therefore, the use of infrared thermography as a tool combined with clinical examination techniques may be a complement to clinical evaluations and support decisions.

KEYWORDS: Infrared Thermography, Biomarkers, Muscle Damage, Endurance, Sport.

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INTRODUCTION

Athletes are constantly exposed to a series of physical overloads through training and matches that cause morphological changes in skeletal muscle, followed by an increase in the concentration of pro-inflammatory and muscle-damaging enzymes (Bishop, Jones & Woods, 2008; Cheng, Jude & Lanner, 2020).

This process can often be accompanied by a reduction in muscular performance and delayed-onset muscle pain (Cheng, Jude & Lanner, 2020). These reactions cause changes in blood flow, which occurs as a result of vasodilation and an increase in the volume of blood circulating in the region, which affects skin temperature (Merla et al, 2010). Controlling heat exchange between the body surface and the external environment plays an important role in regulating body temperature, as it is a way of trying to achieve a balance between metabolic heat production and heat dissipation to the environment (Lim, Byrne & Lee, 2008).

Muscle damage and pain are associated with an increase in the intensity of physical activity, this can be an impactful factor that can weaken the performance and quality during the execution of the athlete's movements (Stewart et al, 2020). Basal skin temperature proves to be a useful indicator related to muscle damage, physiological stress or poor training load adaptation (Priego-Quesada et al, 2019A).

The muscle recovery process is also characterized by an increase in heat in that specific muscle due to the high rate of metabolism and blood flow. (Uchôa et al, 2018). An increase in intensity during training or competition can lead to muscle damage and tissue inflammation (Perez-Guarner et al, 2019).

The use of techniques and tools is important to promote better recovery for the athlete, also preventing the risk of injuries (Carvalho et al, 2021). The interaction

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between the metabolic tools used to detect the increase in biochemical markers, such as creatine kinase (CK), lactate dehydrogenase (LDH), performance analysis through physical tests, the athlete's subjective perception of physical effort and pain, are essential answers for monitoring athletes' training routines (O' Sullivan, O' Sullivan & Gabbett, 2018).

In recent years, Infrared Thermography (IRT) has been gaining more and more space in the sports scene (Carvalho et al, 2021), justified by the ease of diagnosing possible musculoskeletal injuries and even with the objective of preventing them (Côrte & Hernandez, 2016). Infrared thermography is a safe, non-invasive, painless and contactless assessment and diagnosis method (Hildebrandt, Raschner & Ammer, 2010; Uchôa et al, 2018; Côrte et al, 2019; Carvalho et al, 2021), used as a marker of thermal radiation (heat) emitted by the body, and can therefore be used as an instrument to aid in the diagnosis of injuries (Oliveira et al, 2018; Bandeira et al, 2012).

This assessment instrument has been frequently used to identify thermal asymmetries that may in some cases suggest some degree of injury. (Côrte et al. 2019; Gómes-Carmona et al, 2020), which may be mainly associated with inflammation or changes in blood perfusion processes (Hildebrandt, Raschner & Ammer, 2010; Gatt et al, 2015), since athletes are exposed to physical stress and overload during training sessions and competitions that cause changes in blood flow, affecting skin temperature (Merla et al, 2010).

Knowing that injuries trigger inflammatory processes and thus generate heat due to increased local metabolismo (Machado et al, 2009). Then, the inflammatory level can be assessed using temperature gradients (Bandeira et al, 2012). Its monitoring can be interesting for periodizing training loads and preventing injuries

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(Priego-Quesada et al, 2019A). Some researchers are associating the use of physiological markers considered invasive, with other resources such as infrared thermography (Perez-Guarner et al, 2019; Priego-Quesada et al, 2020B; Korman et al, 2021; Rojas-Valverde et al, 2021).

However, if on one hand the serum markers of CK, LDH, among others, indicate the possibility of injury without defining the area of the body, on the other hand, infrared thermography comes as a complement in identifying these regions, as they present a increase in basal temperature. And in this way, this action will highlight a warning for the professionals involved in training this athlete. Therefore, the objective of the present study is to carry out a systematic review to verify whether changes in skin temperature can be compared with the inflammatory physiological changes observed after exercise protocols in trained individuals.

2. METHODS

2.1 Registration and protocol

This systematic review was written in accordance with the PRISMA recommendations demonstrated by Page et al. (2020) and was registered in the International Prospective Register of Systematic Reviews (PROSPERO) under protocol number CRD 42022385463.

2.2 Inclusion/eligibility criteria

In this systematic review, experimental studies that met the following inclusion criteria were included: studies related to sports, with trained athletes, of both sexes and over 18 years of age. A specific time scale was not considered, and any article

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that met the criteria already mentioned in any period of publication could be listed. In the outcome for this systematic review, studies with at least one sport modality were considered; evaluation of biological markers, such as blood lactate and CK; studies that related infrared thermography and biological damage markers; the objective of the study; the protocols used for measurement and the results of the selected studies.

2.3 Search strategy

The search was carried out in January 2023 in the following databases: PubMed, Scielo, Lilacs, Medline, SportDiscus and Embase. The following keywords were used as descriptors for Medical Subject Headings (Mesh), Health Science Descriptors (DeCs) and the Embase Thesaurus (Emtree) or cited in the text: thermography, infrared thermography, skin temperature, infrared image, temperature mapping, CK, blood lactate, biological markers, athletes, sport. The following sentence was obtained using Boolean operators “AND” (among the descriptors (among the synonyms): “Thermography” OR Infrared Thermography” OR “Skin Temperature” OR “Infrared Imaging” OR “Temperature Mapping” AND “Creatine Kinase” OR “Blood Lactate” OR “Biological markers” AND “Athletes” OR “Sport”. Furthermore, no specific language was stipulated for the classification of studies, nor was a publication period determined, meaning any study could be selected of any time and language.

2.4 Study selection

The results were initially selected by titles and abstracts only, and any article that was irrelevant and duplicated would be removed. The researchers then analyzed the remaining articles in full text by two reviewers independently. Any disagreement

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found between the selections was resolved by a third evaluator. Systematic review and/or meta-analysis studies, case studies and abstracts (studies from conference proceedings or those that did not present full text) were excluded.

2.5 Data collection process

The following data were extracted from the selected studies: participants' profile, sex, height, age, fat percentage, sports modality, biological markers referring to muscle damage such as: CK, blood lactate, among others; in addition to the objective of the studies and results.

2.6 Assessment of methodological quality and risk of bias

The methodological quality and risk of bias of the selected studies were evaluated using the QUADAS-2 (Quality Assessment of Diagnostic Accuracy Studies) scale by two experienced evaluators independently. Disagreements were resolved by a third evaluator

The QUADAS-2 tool evaluates the studies included in this systematic review across four main domains that discuss patient selection, index testing, reference standard, flow, and timing. Each domain is assessed in terms of risk of bias, with the first three domains also assessed in terms of their applicability. To assist in the judgment, the questions that the scale refers to are answered as “YES”, “NO” or “UNCERTAIN”. The risk of bias is judged as “LOW”, “HIGH” or “UNCERTAIN”, and have been formulated in such a way that if the answer to all signaling questions for a domain is “yes”, the risk of bias is classified as “low” potential for bias. If any signaling

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question is answered as “no”, the classification is that there is a “high” potential bias. The “uncertain” classification is only used when insufficient data is reported to allow judgment (Whiting et al, 2011).

3. RESULTS

The Figure 1 shows the flowchart of the studies included in this systematic review. A total of 165 studies were extracted from the databases, of which 20 studies were duplicates and were excluded. There were 145 studies left for title and abstract analysis at this first stage. After this stage, 129 studies were excluded. The remaining articles, 16, were considered suitable for full reading and judgment regarding the inclusion criteria. At this stage, eleven articles were excluded and only five studies met the eligibility criteria for this systematic review.

FIGURE 1

Of the studies that aligned with the inclusion criteria, (table 1) highlights the characteristics of the participants who were investigated, such as: number of athletes, sex, age, body weight, height and sport modality practiced. A total of 81 athletes, with an average value of 16.2 participants per study. The largest sample was composed of 21 individuals described by Bandeira and collaborators (2014). However, the smallest sample was evidenced in the study by Fernandes and collaborators (2017), consisting of only 10 athletes. As for sex, the studies analyzed a total of 55 men and 26 women. The sports were also very varied, including: Rugby, Football and Running, both speed and marathon. However, the last modality, running, was the one that showed the greatest advantage in the studies, while three articles related it, the other modalities

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were only present in single studies. However, despite the advantage of the running study, they have particularities, as each article lists a sub-modality of this sport, namely: Sprint running, study proposed by Korman and collaborators (2021), half marathon, evidenced in the research by Perez-Guarner et al, (2019) and Marathon found in Rojas-Valverde (2021). All studies presented mean age, height, body weight and fat percentage, except Bandeira et al (2014), which only presented the mean age. In the study by Fernandes et al (2017), the data not mentioned concerns the body weight of the participants. In the study by Moreira et al (2019), the missing data refers to the percentage of fat of the participating athletes.

TABLE 1

The Table 2 presents the objective of each study, in addition to the basic procedures and main results found by each author.

TABLE 2

Table 3 represents the result of the risk of bias analysis using the QUADAS-2 scale. In this representation, the articles selected for the systematic review were evaluated in four domains, namely: patient selection, index test, reference standard and flow and time. Each domain is assessed for risk of bias as: low, high or uncertain. Assessment also took place for the applicability of each study. However, evaluation occurs in three domains: patient selection, index testing, reference standard. And, all domains were also evaluated following the risk of bias as: low, high or uncertain.

TABLE 3

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In the risk of bias assessment, the study by Bandeira and collaborators (2014) presented a low risk in all domains and, in terms of its applicability. As for the other authors, all were classified as uncertain risk in the reference standard domain. Despite this, regarding applicability, there was a low risk of bias. However, only for Korman et al (2021) regarding the domain for reference standard applicability, this study shows an uncertain risk of bia

4. DISCUSSION

The main objective of this study was to verify whether changes in skin temperature can be compared with the inflammatory physiological changes observed after exercise protocols in trained individuals. Based on the eligibility and selection criteria adopted, as shown in figure 1, the scarcity of studies relating infrared thermography to changes in athletes' biological markers is surprising. Which awakens us to a field of study that still needs to be explored in depth.

Observing the years of publication of the studies selected for this systematic review, we noticed that the oldest study is from 2014 (Bandeira et al), followed later by Fernandes et al, which published later, in 2017. Two years later, Perez -Guarner et al, presents a study in 2019. In 2021, we already see two publications (Korman et al and Rojas-Valverde et al). We believe that this low number of published studies can be explained by the fact that older infrared thermography equipment had low image resolution, high cost, and difficulty in moving the equipment. In addition, there was a lack of methodological standards, limiting the veracity of the studies (Elliot, 1999). However, nowadays this equipment has evolved and has become portable, with high thermal sensitivity and image resolution, which guarantees greater precision in the measurements obtained during the evaluation (Côrte & Hernandez, 2016). In this way,

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this equipment has become an interesting alternative for measuring training load, identifying and preventing injuries, and the proposal for identifying thermoregulatory changes before, during and after exercise (Pinheiro et al, 2021). For Tanda (2016), more modern thermal imaging devices are more suitable for mapping the distribution and evolution of skin temperature during exercise.

The diversity in the results presented among the five studies selected for this systematic review is due to the variety of sports modality explained. In two studies, participating athletes practice team sports, such as Rugby (Bandeira et al, 2014) and Football (Fernandes et al, 2017). The other selected studies present individual characteristics, being: Half-marathon (Perez-Guarner, 2019), Sprinter (Korman et al, 2021) and Marathon (Rojas-valverde, 2021). However, each sport has its own particularities and physiological characteristics.

The increase in temperature identified in studies by infrared thermography in different parts of the body shortly after or days after physical activity determines that blood flow is reestablished to the injured muscle in order to supply immune cells, which makes it necessary for muscle regeneration and with the objective of removing the accumulation of metabolites (Uchôa et al, 2018). Therefore, the results analyzed from the selected studies seem to be in line with the findings of the study by Uchôa et al, (2018) in which they conclude that the muscle recovery process is characterized by increasing heat in that specific muscle that received the demand from the exercise, this, however, happens due to the high rate of metabolism and blood flow (Gatt et al, 2015).

However, in the study proposed by Fernandes and collaborators (2017), skin temperature and CK were moderately related in all ROIs analyzed. And these findings

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revealed that skin temperature is markedly affected and related to the level of inflammation triggered by the football match. However, 48 hours after a football match has not yet been shown to be an efficient amount of time to recover from match-induced muscle damage and concomitant inflammation caused by repeated football matches. Since these athletes are constantly exposed to a series of physical overloads through training and official games that promote physiological changes, such as fatigue, increased concentration of pro-inflammatory enzymes that are harmful to the muscles (Bishop, Jones & Woods, 2008), in addition to increased circulating blood flow in the region (Fernandez-Cuevas et al, 2015). In the study by Rojas-Valverde et al (2021), which aimed to evaluate the effect of basal temperature after a marathon held in an uncontrolled and hot environment, it was concluded that the condition caused an increase in basal temperature 24 hours after the marathon. This result was attributed by the authors to the heat generated by the marathon and its subsequent physiological processes due to the peripheral vasodilation promoted, and this can be reflected in skin temperature. However, muscle damage did not prove to be of great importance, since the study did not obtain a relationship or correlation with biological markers.

However, the study by Perez-Guarner (2019) proved to be different, where during the assessment 24 hours before the half marathon race, the athletes showed an increase in basal temperature, not only that, the half marathon resulted in physiological stress, which which can be seen in the variables analyzed (blood markers, jumping performance and perception of fatigue and pain). However, average skin temperature increased the day before the half marathon race and 48 hours after the race. The explanation given by the authors referring to the increase in basal temperature in the assessment 24 hours before the race is that there was an increase

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in the environmental temperature of the city where the experiment was carried out. However, another possible question is presented by the study by Formenti & Merla (2017), which according to them, there may be activation of the sympathetic activity of the autonomic nervous system.

The research by Korman and collaborators (2021) appears to be somewhat particular, as it relates the basal temperature of the lower limbs with the CK of men and women in a camp of ten consecutive days of training, and analyzed them in two moments, being of morning and night, to observe whether the circadian cycle would be able to interfere with the skin temperature of the lower limbs and CK. As a result, temperature and CK levels changed during each training day and night. In relation to the circadian rhythm, serum CK levels increased during the day and decreased at night. However, basal temperature decreased during the day and increased at night until the sixth day of camp. After this period, the rhythm changed in the opposite way, increasing in the morning and reducing at night. According to the authors, these findings may show that despite skin temperature varying higher or lower, this appears to favorably affect physical performance, due to a greater heat storage capacity, leading us to believe that this phenomenon may indicate a thermal adaptation process (Korman et al, 2021).

The study by Bandeira et al. (2014) presents a particularity: the authors report a certain limitation in the results found, since the measurement of basal temperature and CK occurred only 48 hours after a training session and 48 hours after a game. This makes it difficult to correlate these two techniques, since there is no assessment before the training session and the game, which could clarify the interpretation and show more accurately whether or not serum CK levels and skin temperature increased. This difficulty encountered by the authors in relating skin temperature to

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biomarkers is also shown by the sport analyzed, Rugby. This is a sport where there is diversity in the data, where muscle injuries occur both due to trauma and the intensity of the activities performed. This impacts the time of release and activity of CK, since they depend mainly on the type and intensity of the exercise performed, as described in the study by (Brancaccio, Lippi & Maffulli, 2010). However, not all studies evaluate athletes for a few days as a way of controlling the evolution of basal temperature and biological markers. This can be exemplified in the study by Moreira and collaborators (2019) in which he evaluated 23 judo athletes with a specific test for the modality, and there he evaluated the temperature of 26 regions and blood lactate to compare them with each other, the chosen period it was in the pre-test and 5,10 and 15 minutes after the judo exercise intervention. A similar procedure was found in the research by Tuono and collaborators (2021), where they evaluated the basal temperature of the lower limbs of athletes practicing duathlon sprints at three moments, before the competition, immediately after and 10 minutes after the competition. However, they did not correlate with any type of physiological marker.

Despite this, the low quantity, the divergence of study protocols and the diversity of sports make it difficult to correlate infrared thermography with inflammatory biological markers, triggered in the body after physical activity. Still, this makes additional investigations on this subject even more important and necessary. New studies may add new stages of assessing body temperature and biological markers, including a period before and after training and games.

5. CONCLUSION

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Infrared thermography appears to be a very reliable, safe and non-invasive method for demonstrating the thermal pattern of the skin when used for injury screening and quantifying training loads. However, there are still doubts regarding the effectiveness of infrared thermography in correlating with biological markers of muscle damage. Thus, the use of infrared thermography as a tool combined with clinical examination techniques can be a complement to clinical evaluations and support professionals' decisions. However, there are still many doubts regarding the effectiveness and accuracy of thermal responses in sports, which opens precedents to be explored in future research.

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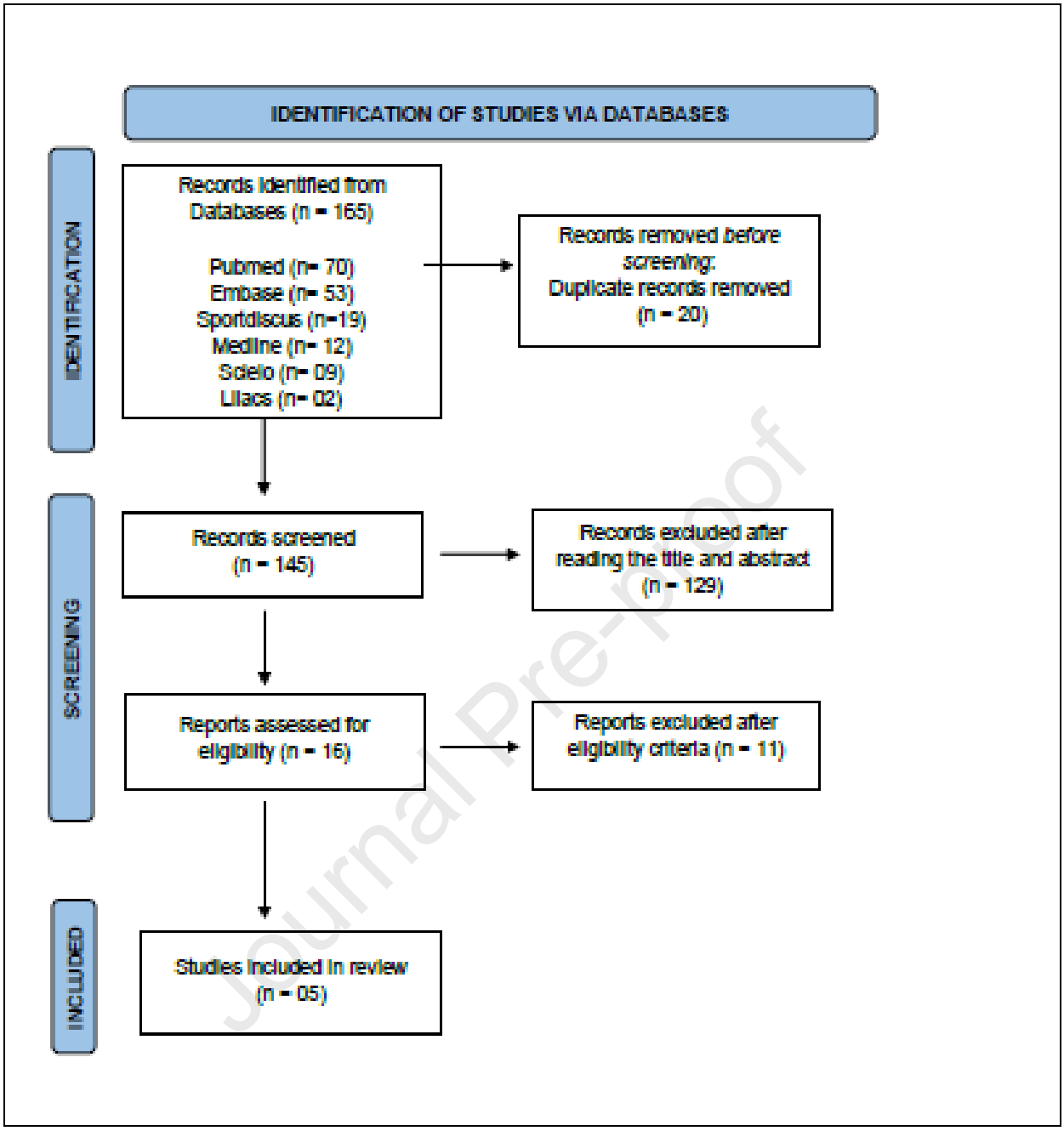

Figure 1: *Flow diagram of the studies included in this systematic review, based on the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) model*

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STUDY	PARTICIPANTS (N)	AGE (YEARS)	WEIGHT (KG)	HEIGHT (CM)	% BODY FAT
Bandeira et al, 2014	21 athletes men Modality: Rugby	±25,06	Not informed	Not informed	Not informed
Fernandes et al, 2017	10 athletes men Modality: Futebol	19 ± 1	Not informed	181,3 ± 6,63	9,2 ± 1,82
Korman et al, 2021	17 athletes (9 men and 8 women) Modality: Runner, Sprinter	26,1± 3,3	84,6 ± 6,8	185,7± 5,7	24,5 ± 0,9
Perez- Guarner et al, 2019	17 athletes (11 men and 6 women) Modality: Running, Half marathon	41 ± 6	66,5 ± 10,0	1,74 ± 0,10	14,0 ± 6,1
Rojas Valverde et al, 2021	16 athletes (9 men and 7 women) Modalitye: Marathon	36 ± 7,3	66,8 ± 12	167 ± 11	21 ± 8

Table 1: *Demonstrates the sample characteristics of each study.*

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AUTHOR YEAR	OBJECTIVE	STUDY PROCEDURE	THERMAL CHARACTERISTIC	MAIN FINDINGS
Bandeira et al, 2014	To analyze the correlation between the variation in skin temperature in different body sites and the variation in CK, at two moments in the activity cycle of rugby athletes.	Blood collection to measure CK; Infrared image 48 hours post training and 48 hours post match. Analysis of 18 ROIs.	Flir A-325, 320x240 pixels of resolution and a thermal sensitivity ≤ 0.08ºC.	There was no correlation between the variation in serum CK level at moments of lowest and highest concentration and the variation in average temperature of the selected area of the muscles analyzed at these same moments.
Fernandes et al, 2017	To examine whether two consecutive football matches would affect skin temperature (Tsk) measured by infrared thermography (IRT) in under-20 football athletes, as well as verify whether Tsk changes would differ between regions of interest.	he study consisted of seven distinct moments: 24 h before the first match (Pre-game); 24 h after the first start (M1-24 h); 48 h after the first start (M1-48 h); 24 h after the second start (M2-24 h); and 48 h after the second start (M2-48 h).	Flir T420, 320x240 pixels of resolution and a thermal sensitivity ≤ 0.05ºC.	The skin temperature of all ROIs was affected by the investigated matches, increasing 24 hours later and decreasing 48 hours later. However, 24 hours after the second match the temperature is still higher in all ROIs, and did not return to pre-match values 48 hours later. CK was significantly higher 48h after the second match compared to 48h after the first match (P= 0.002).
Korman et al, 2021	To analyze the effect of a series of successive speed and power training sessions on simultaneous circadian changes in resting leg skin temperature and blood creatine kinase (CK) levels and to determine the correlation between them.	All participants underwent biochemical and temperature analyzes twice a day, in the morning between 08:00 and 09:00 and between 20:00 and 21:00 local time. On the 10th day (day of departure) only morning analyzes were performed. Each training session consisted of three parts: warm-up, main part and cool-down.	Flir SC 640, 640x480 pixels of resolution and a thermal sensitivity ≤ 0.05ºC.	It was observed a simultaneous increase in blood CK levels and a decrease in morning leg skin temperature. From the first day of the training period, skin temperature in the morning decreased until the 8th day and then stabilized. There was no correlation between temperature and changes in CK level during the multi-day camp.
Perez-Guarner, 2019	Determine the association between skin thermal response and physiological stress after a half Marathon.	Assessments were carried out 48 h before, 24 h before, 24 h after and 48 h after completing the half marathon race. Measurements on each test day included blood markers (creatine kinase [CK] and glutamate oxaloacetate transaminase [GOT]); Perception of pain and fatigue (using a visual analogue scale), skin temperature of upper and lower limb areas; and jumping performance (using the countermovement jump test).	Flir E 60, 320x240 pixels of resolution and a thermal sensitivity ≤ 0.05ºC.	Perceived fatigue and pain increased 24 hours after the half marathon, while jumping performance decreased. No increase in skin temperature was observed in post-competition testing and no regression model was able to predict physiological stress using skin temperature.
Rojas – valverde et al, 2021	To evaluate the effect of a marathon race performed at warm ambient temperatures and analyze the relationship between the basal Tsk response and variations in muscle damage markers.	Skin temperature, lactate and CK were assessed at four moments: 15 days and 45 minutes before the marathon, and 24 hours and 6 days after the Marathon.	Flir T440, 320x240 pixels of resolution and a thermal sensitivity ≤ 0.04ºC.	Tsk was higher the day after the marathon than at the other three times; no relationship or correlation was found between Tsk variation and muscle damage markers ($p > 0.05$).

Table 2: *Description of the objectives, procedures and characteristics of the cameras used in each study, and the main findings of the selected articles.*

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STUDY					APPLICABILITY CONCERNS		
	PATIENTE SELECTION	INDEX TEST	REFERENCE STANDART	FLOW AND TIMING	PATIENT SELECTION	INDEX TEST	REFERENCE STANDART
STUDY 1 Bandeira et al, 2014	Low	Low	Low	Low	Low	Low	Low
STUDY 2 Fernandes et al, 2017	Low	Low	Unclear	Low	Low	Low	Low
STUDY 3 Korman et al, 2021	Low	Low	Unclear	Low	Low	Low	Unclear
STUDY 4 Perez-Guarner et al, 2019	Low	Low	Unclear	Low	Low	Low	Low
STUDY 5 Rojas valverde et al, 2021	Low	Low	Unclear	Low	Low	Low	Low

Table 3: *Methodological quality and risk of bias of selected studies*

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☒The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Not Interests

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